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# THE HINDU-ARABIC NUMERALS

BY

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AND

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*This enhanced digital edition restores the original source passages in Arabic, Devanagari, and Latin manuscript transcription, including the numeral tables of al-Khwarizmi and Fibonacci's Liber Abaci.*

## Preface

So familiar are we with the numerals that bear the misleading name of Arabic, and so extensive is their use in Europe and the Americas, that it is difficult for us to realize that their general acceptance in the transactions of commerce is a matter of only the last four centuries, and that they are unknown to a very large part of the human race to-day.

It seems strange that such a labor-saving device should have struggled for nearly a thousand years after its system of place value was perfected before it replaced such crude notations as the one that the Roman conqueror made substantially universal in Europe. Such, however, is the case, and there is probably no one who has not at least some slight passing interest in the story of this struggle. To the mathematician and the student of civilization the interest is generally a deep one; to the teacher of the elements of knowledge the interest may be less marked, but nevertheless it is real; and even the business man who makes daily use of the curious symbols by which we express the numbers of commerce, cannot fail to have some appreciation for the story of the rise and progress of these tools of his trade.

This story has often been told in part, but it is a long time since any effort has been made to bring together the fragmentary narrations and to set forth the general problem of the origin and development of these numerals. In this little work we have attempted to state the history of these forms in small compass, to place before the student materials for the investigation of the problems involved, and to express as clearly as possible the results of the labors of scholars who have studied the subject in different parts of the world. We have had no theory to exploit, for the history of mathematics has seen too much of this tendency already, but as far as possible we have weighed the testimony and have set forth what seem to be the reasonable conclusions from the evidence at hand.

To facilitate the work of students an index has been prepared which we hope may be serviceable. In this the names of authors appear only when some use has been made of their opinions or when their works are first mentioned in full in a footnote.

If this work shall show more clearly the value of our number system, and shall make the study of mathematics seem more real to the teacher and student, and shall offer material for interesting some pupil more fully in his work with numbers, the authors will feel that the considerable labor involved in its preparation has not been in vain.

We desire to acknowledge our especial indebtedness to Professor Alexander Ziwet for reading all the proof, as well as for the digest of a Russian work, to Professor Clarence L. Meader for Sanskrit transliterations, and to Mr. Steven T. Byington for Arabic transliterations and the scheme of pronunciation of Oriental names, and also our indebtedness to other scholars in Oriental learning for information.

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## Pronunciation of Oriental Names

(S) = in Sanskrit names and words; (A) = in Arabic names and words.

**b**, **d**, **f**, **g**, **h**, **j**, **l**, **m**, **n**, **p**, **sh (A)**, **t**, **th (A)**, **v**, **w**, **x**, **z**, as in English.

**a**, (S) like *u* in *but*: thus pandit, pronounced pundit. (A) like *a* in *ask* or in *man*.

**a with macron**, as in father.

**c**, (S) like *ch* in *church* (Italian *c* in *cento*).

**d with dot below**, **n with dot below**, **s with dot below**, **t with dot below**, (S) made with the tip of the tongue turned up and back into the dome of the palate.

**d with dot below**, **s with dot below**, **t with dot below**, **z with dot below**, (A) made with the tongue spread so that the sounds are produced largely against the side teeth.

**d with dot below (A)**, like *th* in *this*.

**e**, (S) as in *they*. (A) as in *bed*.

**g with dot below**, (A) a voiced consonant formed below the vocal cords; its sound is compared by some to a *g*, by others to a guttural *r*.

**i**, as in *pin*. **i with macron**, as in *pique*.

**kh**, (A) the hard *ch* of Scotch *loch*, German *ach*.

**m**, **n**, (S) like French final *m* or *n*, nasalizing the preceding vowel.

**n with dot above**, like *ng* in singing.

**o**, (S) as in *so*. (A) as in *obey*.

**q**, (A) like *k* (or *c*) in *cook*; further back in the mouth.

**r**, (S) English *r*, smooth and untrilled. (A) stronger.

**s with acute**, (S) English *sh* (German *sch*).

**u**, as in *put*. **u with macron**, as in *rule*.

**ayin / hamza**, (A) a sound kindred to the spiritus lenis.

**Accent**: (S) as if Latin; (A) on the last syllable that contains a long vowel or a vowel followed by two consonants.

# 1 Early Ideas of Their Origin

It has long been recognized that the common numerals used in daily life are of comparatively recent origin. The number of systems of notation employed before the Christian era was about the same as the number of written languages, and in some cases a single language had several systems. The Egyptians, for example, had three systems of writing, with a numerical notation for each; the Greeks had two well-defined sets of numerals, and the Roman symbols for number changed more or less from century to century. Even to-day the number of methods of expressing numerical concepts is much greater than one would believe before making a study of the subject, for the idea that our common numerals are universal is far from being correct.

It will be well, then, to think of the numerals that we still commonly call Arabic, as only one of many systems in use just before the Christian era. As it then existed the system was no better than many others, it was of late origin, it contained no zero, it was cumbersome and little used, and it had no particular promise. Not until centuries later did the system have any standing in the world of business and science; and had the place value which now characterizes it, and which requires a zero, been worked out in Greece, we might have been using Greek numerals to-day instead of the ones with which we are familiar.

Of the first number forms that the world used this is not the place to speak. Many of them are interesting, but none had much scientific value. In Europe the invention of notation was generally assigned to the eastern shores of the Mediterranean until the critical period of about a century ago,—sometimes to the Hebrews, sometimes to the Egyptians, but more often to the early trading Phoenicians.

The idea that our common numerals are Arabic in origin is not an old one. The mediaeval and Renaissance writers generally recognized them as Indian, and many of them expressly stated that they were of Hindu origin. Others argued that they were probably invented by the Chaldeans or the Jews because they increased in value from right to left, an argument that would apply quite as well to the Roman and Greek systems, or to any other. It was, indeed, to the general idea of notation that many of these writers referred, as is evident from the words of England's earliest arithmetical textbook-maker, Robert Recorde (c. 1542):

> "In that thinge all men do agree, that the Chaldays, whiche fyrste inuented thys arte, did set these figures as thei set all their letters. for they wryte backwarde as you tearme it, and so doo they reade. And that may appeare in all Hebrewe, Chaldaye and Arabike bookes...where as the Greekes, Latines, and all nations of Europe, do wryte and reade from the lefte hand towarde the ryghte."

Others, and among them such influential writers as Tartaglia in Italy and Kobel in Germany, asserted the Arabic origin of the numerals, while still others left the matter undecided or simply dismissed them as "barbaric." Of course the Arabs themselves never laid claim to the invention, always recognizing their indebtedness to the Hindus both for the numeral forms and for the distinguishing feature of place value.

Foremost among these writers was the great master of the golden age of Bagdad, one of the first of the Arab writers to collect the mathematical classics of both the East and the West, preserving them and finally passing them on to awakening Europe. This man was Mohammed the Son of Moses, from Khowarezm, or, more after the manner of the Arab, **Mohammed ibn Musa al-Khowarazmi**,[1] a man of great learning and one to whom the world is much indebted for its present knowledge of algebra and of arithmetic.

[1] His full name was Abu 'Abdallah Mohammed ibn Musa al-Khowarazmi. He was born in Khowarezm, "the lowlands," the country about the present Khiva and bordering on the Oxus, and lived at Bagdad under the caliph al-Mamun. He died probably between 220 and 230 of the Mohammedan era, that is, between 835 and 845 a.d.

Of him there will often be occasion to speak; and in the arithmetic which he wrote, and of which Adelhard of Bath (c. 1130) may have made the translation or paraphrase, he stated distinctly that the numerals were due to the Hindus. This is as plainly asserted by later Arab writers, even to the present day. Indeed the phrase علم هندي (*'ilm hindi*), "Indian science," is used by them for arithmetic, as also the adjective هندي (*hindi*) alone.

Probably the most striking testimony from Arabic sources is that given by the Arabic traveler and scholar **Mohammed ibn Ahmed**, **Abu 'l-Rihan al-Biruni** (973–1048), who spent many years in Hindustan. He wrote a large work on India, the "Book of the Ciphers," unfortunately lost, which treated doubtless of the Hindu art of calculating, and was the author of numerous other works. Al-Biruni was a man of unusual attainments, being versed in Arabic, Persian, Sanskrit, Hebrew, and Syriac, as well as in astronomy, chronology, and mathematics.

In his work on India he gives detailed information concerning the language and customs of the people of that country, and states explicitly that the Hindus of his time did not use the letters of their alphabet for numerical notation, as the Arabs did. He also states that the numeral signs called *ahka*[2] had different shapes in various parts of India, as was the case with the letters.

Preceding Al-Biruni there was another Arabic writer of the tenth century, Motahhar ibn Tahir, author of the Book of the Creation and of History, who gave as a curiosity, in Indian (Nagari) symbols, a large number asserted by the people of India to represent the duration of the world.

Mention should also be made of a widely-traveled student, **Al-Mas'udi** (885?–956), whose journeys carried him from Bagdad to Persia, India, Ceylon, and even across the China sea. He states that the wise men of India, assembled by the king, composed the *Sindhind*. Further on he states, upon the authority of the historian Mohammed ibn 'Ali 'Abdi, that by order of Al-Mansur many works of science and astrology were translated into Arabic, notably the Sindhind (Siddhanta).

The oldest work, in any sense complete, on the history of Arabic literature and history is the *Kitab al-Fihrist*, written in the year 987 a.d., by Ibn Abi Ya'qub al-Nadim. It is of fundamental importance for the history of Arabic culture.

Leonard of Pisa, of whom we shall speak at length in the chapter on the Introduction of the Numerals into Europe, wrote his *Liber Abbaci* in 1202. In this work he refers frequently to the nine Indian figures, thus showing again the general consensus of opinion in the Middle Ages that the numerals were of Hindu origin. Some interest also attaches to the oldest documents on arithmetic in our own language. One of the earliest treatises on algorism is a commentary on a set of verses called the *Carmen de Algorismo*, written by Alexander de Villa Dei (Alexandre de Ville-Dieu), a Minorite monk of about 1240 a.d.

[2]The Hindu name for the symbols of the decimal place system.

## 2 Hindu-Arabic Systems Used by the Arabs

When the Arabs, carried on the wave of religious enthusiasm, had spread their conquests from the Atlantic to the Indus, they found everywhere the use of the Greek alphabetic numerals. In Syria they adopted the Greek system, in Egypt the Coptic, and in Persia the system used by the Persians. These systems were, however, of little value for mercantile purposes, and the Arabs early saw the advantages of the Hindu system which had reached them through the translations of the astronomical works of Sindhind.

The spread of the numerals among the Arabs was slow. In the East, under the Abbasid caliphs of Bagdad, the Hindu system found ready acceptance among scholars, but the common people and the merchants continued to use their old systems. It was not until the eleventh century that the Hindu numerals became common in Arabic works, and even then they were used chiefly by mathematicians and astronomers.

The earliest known Arabic work containing the Hindu numerals is the arithmetic of Al-Khowarazmi, already mentioned. This work, which bore the title *Algoritmi de numero Indorum*, was translated into Latin, probably by Adelhard of Bath, in the twelfth century. The Arabic original is lost, but several Latin versions exist, and from these we know that Al-Khowarazmi gave the nine Hindu numerals and explained their use in addition, subtraction, multiplication, and division.

The numerals as used by the Arabs underwent certain modifications. The forms used in the Eastern part of the Arab empire, centering in Bagdad, were different from those used in the Western part, centering in Cairo and later in Spain. The Eastern forms were called *Indien* (Indian), while the Western forms were called *gobar* (dust).

### The al-Khwarizmi Numeral Table

The following table reconstructs the numeral forms as given by al-Khwarizmi in his *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning), as transmitted through the Latin manuscripts of the twelfth century.

Table 1: Al-Khwarizmi's Hindu Numerals (reconstructed from Latin MSS)

| **Value** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Latin MS form (Bonn, 12th c.)** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| **Eastern Arabic (10th c.)** | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| **Modern Arabic** | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

### Eastern and Western Arabic Numeral Forms

The gobar numerals themselves were first made known to modern scholars by Silvestre de Sacy, who discovered them in an Arabic manuscript from the library of the ancient abbey of St.-Germain-des-Pres. The system has nine characters, but no zero. A dot above a character indicates tens, two dots hundreds, and so on.

In form these gobar numerals resemble our own much more closely than the Arab numerals do. They varied more or less, but were substantially as follows.

Table 2: Eastern vs. Western (Gobar) Arabic Numerals

| **Value** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Eastern Arabic (Bagdad)** | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| **Western Arabic / Gobar** | . | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| **Maghribi (Spain/N. Africa)** | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

Among the Arab writers who used and spread the Hindu numerals may be mentioned Al-Kindi (800–870), Al-Sufi (died 986), Al-Biruni (973–1048), and Al-Hassar (c. 1150). Al-Kindi wrote several works on arithmetic and the use of the Indian method of reckoning. Al-Sufi wrote a large work on arithmetic. Al-Biruni, as already mentioned, was a great traveler and scholar who used the Hindu numerals extensively in his astronomical and mathematical works. Al-Hassar wrote a work on arithmetic in which the second chapter was "On the dust figures."

The Arabic system of numeration, like the Hindu, was based on the principle of place value. The Arabs adopted the zero, which they called *as-sifr* or *sifr*, and used it in the same way as the Hindus.

Italian *zefiro*, Spanish *cero*, Portuguese *zero*

French *zero*, English *zero*, German *Null*

Medieval Latin *zephirum* (Fibonacci, 1202)

*Source: See F. Woepcke, Sur l'introduction de l'arithmetique indienne en Occident, Rome, 1859*

These methods were known as *hisab al-hind*, the Indian reckoning, and *hisab al-gobar*, the dust reckoning. The spread of the numerals among the Arabs was thus a gradual process, extending over several centuries. It was aided by the great intellectual activity that characterized the Arab world from the ninth to the twelfth century.

## 3 Early Hindu Forms

While it is generally conceded that the scientific development of astronomy among the Hindus towards the beginning of the Christian era rested upon Greek or Chinese sources, yet their ancient literature testifies to a high state of civilization, and to a considerable advance in sciences, in philosophy, and along literary lines, long before the golden age of Greece.

From the earliest times even up to the present day the Hindu has been wont to put his thought into rhythmic form. The first of this poetry—it well deserves this name, being also worthy from a metaphysical point of view—consists of the Vedas, hymns of praise and poems of worship, collected during the Vedic period which dates from approximately 2000 B.C. to 1400 B.C.

Following this work, or possibly contemporary with it, is the Brahmanic literature, which is partly ritualistic (the Brahmanas), and partly philosophical (the Upanishads). Our especial interest is in the Sutras, versified abridgments of the ritual and of ceremonial rules, which contain considerable geometric material used in connection with altar construction.

### The Brahmi and Kharosthi Numerals

The early Hindu numerals may be classified into three great groups, (1) the Kharosthi, (2) the Brahmi, and (3) the word and letter forms; and these will be considered in order.

The Kharosthi numerals are found in inscriptions formerly known as Bactrian, Indo-Bactrian, and Aryan, and appearing in ancient Gandhara, now eastern Afghanistan and northern Punjab. The alphabet of the language is found in inscriptions dating from the fourth century B.C. to the third century A.D., and from the fact that the words are written from right to left it is assumed to be of Semitic origin.

Not until the time of the powerful King Asoka, in the third century B.C., do numerals appear in any inscriptions thus far discovered; and then only in the primitive form of marks. These Asoka inscriptions, some thirty in all, are found in widely separated parts of India.

### The Nana Ghat Numerals (2nd c. B.C.)

The next very noteworthy evidence of the numerals, and this quite complete as will be seen, is found in certain cave inscriptions dating back to the first or second century A.D. In these, the Nasik cave inscriptions, the forms are as follows.

१ २ ३ ४ ५ ६ ७ ८ ९

1 2 3 4 5 6 7 8 9

The Nana Ghat cave inscriptions (c. 150 B.C.) contain the earliest known Brahmi numeral system. These are vertical and perpendicular forms, quite distinct from the later cursive Devanagari. The numeral for 10 is a simple cross-like mark (X), and the numbers 20–90 are formed by ligatures.

*Source: See G. Buhler, Indische Palaeographie, Strassburg, 1896, Tafel III; J. F. Fleet, Corpus Inscriptionum Indicarum, Vol. III*

Table 3: Development of Brahmi Numerals

| Source | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Asoka (3rd c. B.C.)** | I | II | III | + | / | | | | | X |
| **Nana Ghat (2nd c. B.C.)** | – | = | ≡ | +/ | $h$ | 6 | 7 | 8 | ?0 | X |
| **Nasik Caves (1st c. A.D.)** | – | = | ≡ | ⊕ | $h$ | 6 | 7 | 8 | ?0 | ) |
| **Kushana (1st–3rd c.)** | – | = | ≡ | ⊕ | $h$ | 6 | 7 | 8 | ?0 | ) |

## The Development of Devanagari Numerals

From this time on, until the decimal system finally adopted the first nine characters and replaced the rest of the Brahmi notation by adding the zero, the progress of these forms is well marked.

With respect to these numerals it should first be noted that no zero appears in the table, and as a matter of fact none existed in any of the cases cited. It was therefore impossible to have any place value, and the numbers like twenty, thirty, and other multiples of ten, one hundred, and so on, required separate symbols except where they were written out in words.

० १ २ ३ ४ ५ ६ ७ ८ ९

0 1 2 3 4 5 6 7 8 9

The modern Devanagari numerals, which developed from the Brahmi forms through the Gupta and Nagari periods. The zero is called शून्य (*sunyabindu*, later shortened to *sunya*), meaning "void" or "empty."

*Source: See Buhler, Indische Palaeographie; Thibaut, Astronomie, Astrologie und Mathematik*

The ancient Hindus had no less than twenty of these symbols, a number that was afterward greatly increased. Having now examined types of the early forms it is appropriate to turn our attention to the question of their origin. As to the first three there is no question. The I or — is simply one stroke, or one stick laid down by the computer. The II represents two strokes or two sticks, and so for the III and IV.

From some primitive form came the two of Egypt, of Rome, of early Greece, and of various other civilizations. From some primitive form — came the Chinese symbol, which is practically identical with the symbols found commonly in India from 150 B.C. to 700 A.D.

## The Word and Letter Numeral Systems

Mention should also be made, in this connection, of a curious system of alphabetic numerals that sprang up in southern India. In this we have the numerals represented by the letters as given in the following table. By this plan a numeral might be represented by any one of several letters, and thus it could the more easily be formed into a word for mnemonic purposes.

क ख ग घ ङ च छ ज झ
1 2 3 4 5 6 7 8 9

ट ठ ड ढ ण त थ द ध
1 2 3 4 5 6 7 8 9

प फ ब भ म
1 2 3 4 5

The Katapayadi system of alphabetic numerals from southern India. For example, the word भवतिः (*bhavatih*) has the value 4+4+6 = 14, read from right to left.

*Source: See A. C. Burnell, South Indian Palaeography, 2nd ed., London, 1878*

# 4 Introduction into Europe

## The Definite Introduction: Fibonacci's *Liber Abaci*

The definite introduction of the Hindu-Arabic numerals into Europe is usually attributed to Leonardo of Pisa, known as Fibonacci, who wrote his famous *Liber Abaci* in 1202. Leonardo had traveled extensively in the East, visiting Egypt, Syria, Greece, and Sicily, and had learned the Hindu methods of reckoning from Arab teachers. In his book he explained the use of the nine Indian figures and the zero, and showed how calculations could be performed with them.

The *Liber Abaci* was not, however, the first work in Europe to mention the Hindu numerals. As we have seen, there is evidence that the numerals were known in Spain and possibly in other parts of Europe as early as the tenth century. But Leonardo's work was the first to give a systematic and comprehensive treatment of the new system.

9 8 7 6 5 4 3 2 1

*Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur.*

"With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below."

*Source: Boncompagni, op. cit., p. 2*

The reception of the numerals in Europe was slow. The medieval universities, with their devotion to the classics, were at first hostile to the new system. An important figure in the early history of the numerals in Europe was Gerbert, who became Pope Sylvester II in 999. Gerbert had studied in Spain and had learned the numerals there.

## The Carmen de Algorismo and Medieval Latin Texts

Among the early European works on arithmetic using the Hindu numerals may be mentioned the *Carmen de Algorismo* of Alexander de Villa Dei (c. 1240), the *Algorismus* of Johannes de Sacrobosco (c. 1250).

The spread of the numerals from Italy to the rest of Europe was a gradual process. France and Germany adopted them in the fifteenth century, and England in the sixteenth. Even then, the Roman numerals continued to be used for many purposes, and it was not until the eighteenth century that the Hindu-Arabic numerals became completely dominant in Europe.

# 5 Numerals in the East

The numerals of the Hindu-Arabic system spread not only westward to Europe but also eastward through Persia, Central Asia, and into China. In each region they underwent modifications adapted to local scripts and computational traditions.

## Persian and Central Asian Forms

In Persia, the Arabic numerals were adopted with minor modifications. The Persian forms of the numerals, which are still used today in Iran, Afghanistan, and parts of Central Asia, are derived from the Eastern Arabic forms but with certain calligraphic adaptations to the Persian (Nasta'liq) script tradition.

Table 4: Persian Numeral Forms

| **Value** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Persian (Modern)** | ۰ | ۱ | ۲ | ۳ | ۴ | ۵ | ٦ | ۷ | ۸ | ۹ |

## Forms in India and Southeast Asia

The numerals also continued to develop independently in India. The modern Devanagari forms used in India and Nepal are direct descendants of the Brahmi numerals, parallel to but distinct from the line of development that led through the Arabs to Europe.

० १ २ ३ ४ ५ ६ ७ ८ ९

0 1 2 3 4 5 6 7 8 9

The Devanagari numerals shown above are used throughout northern India, Nepal, and in Sanskrit and Hindi texts. Regional variants exist in Bengali, Gujarati, Gurmukhi, Kannada, Malayalam, Tamil, and Telugu scripts.

*Source: See Burnell, South Indian Palaeography; Enthoven, Numeral Notation in India*

## The Eastern Spread to China

In China, the numerals were known through Arab and Persian merchants, but the Chinese already possessed their own well-developed system of rod numerals and character-based numeration. The so-called "Arabic" numerals were not adopted in China until the modern period, though they were known to Chinese mathematicians through Islamic astronomical treatises.

The Chinese system of rod numerals, which dates from at least the Han dynasty (206 B.C.–220 A.D.), used horizontal and vertical strokes in a place-value system, and this system already embodied the essential principle of positional notation. The Chinese had, in effect, a decimal place-value system without a written zero symbol—they used a blank space instead, much as the Babylonians had done with their sexagesimal system.

## Special Feature: Complete Numeral Comparison Table

The following comprehensive table presents the Hindu-Arabic numerals in their various historical forms, from the earliest Brahmi inscriptions through to modern European usage. This table allows the reader to trace the evolution of each digit across civilizations.

Table 5: Complete Historical Comparison of Numeral Forms

| Source / Period | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Brahmi (3rd c. B.C.) | — | I | II | III | + | $h$ | | | | X |
| Nana Ghat (2nd c. B.C.) | — | — | = | ≡ | +/ | $h$ | 6 | 7 | ?0 | ) |
| Nasik Caves (1st c. A.D.) | — | — | = | ≡ | ⊕ | $h$ | 6 | 7 | ?0 | ) |
| Gupta (4th–6th c.) | • | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Devanagari | ० | १ | २ | ३ | ४ | ५ | ६ | ७ | ८ | ९ |
| Eastern Arabic | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| Western Arabic (Gobar) | . | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Maghribi (Spain) | ◯ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Leonardo of Pisa (1202) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 15th c. European | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Modern European | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

## Special Feature: The al-Khwarizmi Numeral Table

Al-Khwarizmi's *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning) is the earliest systematic Arabic treatise on the Hindu numeral system. Although the Arabic original is lost, we possess Latin translations made in the twelfth century, notably the manuscript preserved at the University of Cambridge (MS Ii.6.5) and fragments in other collections.

The following table reconstructs al-Khwarizmi's numeral system from the Latin manuscripts and compares it with the Eastern Arabic forms preserved in later Arabic astronomical tables.

Table 6: The al-Khwarizmi Numeral System (Reconstructed)

| Value | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| *Dixit Algorizmi* (Cambridge MS) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Eastern Arabic (10th c.) | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| Modern Arabic | ٠ | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

The Latin translator rendered the Arabic title as *Dixit Algorizmi* ("Al-Khwarizmi said"), whence comes our word "algorithm." The work systematically explains:

1. The nine Hindu figures and the zero
2. The principles of place value
3. Operations: addition, subtraction, doubling, halving
4. Multiplication and division
5. Extraction of square roots

## Special Feature: Fibonacci's *Liber Abaci* Passages in Latin

The following passages from Leonardo of Pisa's *Liber Abaci* (1202) are among the earliest extant Latin texts describing the Hindu-Arabic numeral system in Europe. The manuscript tradition is based on the edition of Baldassarre Boncompagni (1857), prepared from the Florence manuscript (Biblioteca Laurenziana, Conv. Sopp. 261).

9 8 7 6 5 4 3 2 1

*Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur. Nam unitas in primo gradu, id est in ultima figura, posita, ipsam representat. Si uero in secundo, ipsam representat, et in primo loco cifram ponit. Et si in tertio gradu posita fuerit, ipsam representat, et in alio cifras ponit. Similiter intelligendum est de omnibus aliis figuris, quamlibet in quolibet gradu positam representare.*

"The nine Indian figures are these: 9, 8, 7, 6, 5, 4, 3, 2, 1. With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below. For unity placed in the first grade, that is in the last figure, represents itself. If indeed in the second, it represents itself and places a zero in the first place. And if it is placed in the third grade, it represents itself and places zeros in the other places. Similarly it is to be understood concerning all other figures, that any figure placed in any grade represents itself."

*Source: Boncompagni, op. cit., pp. 2–3*

## Conclusion

In conclusion, we may say that the history of the Hindu-Arabic numerals is a long and complex one, stretching over many centuries and involving many different civilizations. From their earliest beginnings in India, through their development by the Arabs, to their final adoption in Europe, these numerals have had a remarkable career. They are one of the most important inventions in the history of civilization, and their influence on science, commerce, and daily life can hardly be overestimated.

The study of their history reveals the deep connections that have always existed between the peoples of the East and the West, and shows how knowledge has been transmitted from one civilization to another along the great trade routes that have linked the world since ancient times. It is a story that deserves to be known by all who use these symbols, which have become so familiar that we are apt to forget how much labor and ingenuity went into their creation.

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End of the Enhanced Edition

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